

Capstone: ResViz

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Client brief

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Project description

This project is an interactive, web-based educational tool designed to visualize resolution. You will build a web application where users input logical formulas and the app shows how a resolution-based algorithm searches for contradictions to prove or disprove satisfiability. You will develop a Python back-end to handle tokenizing, parsing, and general processing of custom logical expressions. You are expected to produce a reusable API, which will then be consumed by a front-end, that makes use of a modern framework like React or Vue, to visualize the algorithm step-by-step.

Client background

- ▶ For this project, I am a trainer for a 'Learner admissions' committee at an educational institution (e.g., <https://www.westerncape.gov.za/education/service/learner-admissions>)
- ▶ Committee members are trained to formalise facts about each student using propositional logic
- ▶ They determine the satisfiability of facts using resolution
 - ▶ Process: compile a KB, convert it to CNF, and then apply resolution to the set of clauses
- ▶ This is currently done by pen and paper.

Challenges and opportunities

- ▶ Replace pen and paper with a webapp
 - ▶ We'd like to quickly type propositional formulas
- ▶ Members often forget what each proposition means (so there is a need for ease of annotation)
- ▶ We'd like to be able to trace how the solution/answer was reached for purposes of explainability
- ▶ Resolution algo impl. should not blindly compare all clauses. Need for prioritisation of clauses 'related' to the goal

Challenges and opportunities

- ▶ The webapp should be split into backend/frontend.
- ▶ There is no existing API at the moment
- ▶ There is no agreed upon domain-specific language to use for specifying facts
- ▶ There are no guidelines regarding the best way to visualise the algorithm
- ▶ There is an existing Python-based propositional logic parser, CNF convertor, and resolution algo. implementation (some of these resources may be provided)
- ▶ There is no limitation regarding the re-writing of some of the code that may be provided